Welfare Dimension Boundary and Interaction Protocol

v1.4 Current Release Integration

Release: MathGov Core Release 2026.09 v12.4 / SGP v8.3 - Calculability, Type-Integrity, and Cross-Artifact Synchronization Public Research Source Release

Exact current companion pins: Canon v12.4; SGP v8.3; ripple.md v5.3; Agent System v12.2; CSV v2.2; Cascade v2.4; Reproducibility v1.2; WDBIP v1.4; RLS Validation v2.4; Primer v4.2; Public Introduction v12.4; PC-AEP/MFDI/Source-Coupling v2.1; Aligners Sheet v5.4.

Integrity surface	Current v12.4 requirement	Claim boundary
Role boundary	WDBIP governs construct boundaries, evidence cards, allocation, interaction, temporal compatibility, subgroup visibility, and uncertainty beneath RLS.	It is not an eighth welfare dimension, an option gate, or a moral-status instrument.
Machine line	The current schema, example, validator, and vectors use the v1_4 namespace and require a PCC locator where applicable.	Schema passage is not construct validation.
Maturity	Empirical reliability, validity, invariance, and factor structure remain provisional and must be tested in preregistered studies.	No latent structure is asserted by document design alone.

WDBIP v1.4 - A Welfare Measurement and Conformance Protocol Beneath RippleLogic Scoring

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Framework: MathGov
Decision architecture: RippleLogic v12.4
Related moral-status interface: SGP v8.3
Protocol role: Companion welfare-measurement and record-conformance protocol beneath RLS

Status: Public research-source measurement protocol and conformance standard. The record rules are implementable; the seven-dimension measurement architecture remains empirically provisional. Not a new gate, not a new welfare dimension, and not empirical validation.
Release focus: Dimension-set migration, boundary reliability, allocate-versus-split discipline, time and subgroup integrity, weight sensitivity, psychometric operationalization, runnable validation planning, and core-interface readiness.

> **Measure the parts without pretending they are isolated. Measure the relationships without erasing the parts.**

Contents

Source-boundary and integration-status rule

If this protocol conflicts with the released RippleLogic Canon, the Canon controls. WDBIP governs only the classification, evidence documentation, interaction mapping, and validation of welfare-dimension inputs used by RLS. It does not alter the canonical cascade, rights coverage, TRC, CSV, RLS formula, Union Scopes, constitutional weights, or lawful authority requirements.

WDBIP v1.4 is release-complete as a standalone companion package. It is a reciprocally registered MathGov normative implementation companion because the exact WDBIP version and hash are listed in the governing Core Component Map, Source Hierarchy, Canon specification contract, and package manifest. A prose claim of integration does not substitute for those synchronized artifacts. The v12.4 release incorporates the controlling Canon and SGP interface inserts. WDBIP remains subordinate to those governing sources.

The governing architecture remains:

RG -> RF/NCRC -> TRC -> CSV -> RLS

WDBIP is **not an eighth welfare dimension** and **not a sixth gate**. It operates inside the construction and review of RLS welfare evidence after qualification requirements are respected. If WDBIP reveals a previously omitted rights, catastrophe, structural-viability, physical-admissibility, or authority issue, the run reopens the relevant existing layer. WDBIP does not decide selectability by itself.

Executive standard

MathGov uses seven welfare dimensions as a public anti-omission map:

1. D1 Material
2. D2 Health
3. D3 Social
4. D4 Knowledge
5. D5 Agency
6. D6 Meaning
7. D7 Environment

These dimensions are related but not interchangeable. Health can be affected by material insecurity, social isolation, misinformation, loss of agency, meaning distress, and environmental exposure. That does not make those conditions identical to health. A scientifically useful system must preserve construct boundaries while recording causal, enabling, moderating, and feedback relationships.

Typed 7×7 field clarification. For compatibility with the Canon, D1-D7 retain the stable name Welfare Dimensions; however, the full Union-Scope × Dimension matrix is interpreted as a multidimensional accountability-and-flourishing field rather than 49 uniform phenomenally experienced welfare states. A cell or effect token may represent DIRECT_WELFARE, COLLECTIVE_FUNCTIONING, ENABLING_CONDITION, STRUCTURAL_INTEGRITY, ECOLOGICAL_CONDITION, EPISTEMIC_OBSERVATION_CAPACITY, or SYMBOLIC_MEANING_CONDITION. The record MUST state the applicable interpretation when ambiguity is material, and it MUST NOT imply that a non-sentient system experiences welfare without admissible SGP/evidence support.

WDBIP therefore requires:

0. **Dimension-set pin:** every record declares the governing seven-dimension taxonomy version. Existing records remain immutable if a future governed revision changes the taxonomy.
1. **Primary dimension:** each material effect token has one primary welfare dimension for a declared bearer, baseline, and time window.
2. **Canonical scope treatment:** each token records its represented Union Scopes and inherits the Canon's redundancy rules. DEDUPLICATION requires a declared primary scope, ALLOCATION requires bounded coefficients, and EMERGENT_SCALE_JUSTIFICATION requires genuinely distinct scale-specific states.
3. **Profile before index:** each dimension is represented as an evidence profile before any scalar summary is attempted.
4. **Token-level pathways:** cross-dimensional and cross-scope consequences are recorded between bounded effect tokens rather than silently copied into several cells.
5. **Dependence control:** shared evidence, mediators, repeated outcomes, correlated indicators, and scope duplication are disclosed to prevent inflation.
6. **Uncertainty discipline:** unknown, contested, culturally unstable, and weakly measured states are not converted into zero or false precision.
7. **PCC attachment:** every claim-bearing WDBIP record is attached to or hash-referenced from the run's PCC.
8. **Gate feedback:** newly discovered gate-material hazards reopen RG, RF/NCRC, TRC, or CSV.
9. **Empirical revisability:** dimensions and instruments must survive reliability, discriminant-validity, invariance, predictive-value, and decision-utility testing.

1. Purpose and problem

1.1 Purpose

WDBIP provides a reproducible method for answering seven questions:

1. Which welfare dimension primarily contains a claimed effect?
2. Which Union Scope treatment governs the token under the Canon's multi-scope and redundancy rules?
3. Which other dimensions or scopes are affected through material pathways?
4. How should those pathways be evidenced without double counting?
5. How does the record attach to the run's PCC and, where relevant, SGP record?
6. What may be measured about meaning, spiritual experience, health, and environmental conditions without exceeding the evidence?
7. What evidence would weaken, revise, merge, divide, or reject a dimension or pathway claim?

1.2 The category-collapse problem

Multidimensional welfare systems face two opposite errors.

Isolation error: dimensions are treated as sealed boxes, so material pathways disappear. For example, degraded air quality is recorded under Environment while respiratory morbidity is omitted from Health.

Collapse error: every determinant and consequence is placed inside one broad category, often Health or Flourishing. The result becomes analytically unusable because material insecurity, social belonging, agency, meaning, and ecological integrity lose their distinct causal and normative roles.

WDBIP rejects both errors:

Distinct dimensions + typed interactions != isolated silos or one undifferentiated score.

1.3 Why this matters for RLS

RLS ranks only selectable options using residual welfare evidence. A numerical cell value is not evidence by itself. If the underlying effect is placed in the wrong dimension, counted repeatedly, supported by a culturally invalid instrument, or assigned false causal meaning, the final score may be arithmetically correct but substantively invalid.

WDBIP improves the evidence surface feeding RLS. It does not make RLS a truth machine.

2. Authority, placement, and non-claims

2.1 Placement beneath RLS

WDBIP sits between evidence construction and residual welfare scoring:

Stage	Function	WDBIP role
RG	Grounds claims and evidence boundaries	Receives only evidence allowed by RG; returns category or evidence failures for narrowing or escalation.
RF/NCRC	Protects non-compensatory rights	Does not redefine rights; forwards severe rights evidence discovered during welfare analysis.
TRC	Bounds catastrophe and irreversible ruin	Does not absorb tail risk; forwards newly identified catastrophe pathways.
CSV	Tests containment and structural viability	Forwards dependency, ecological, implementation, monitoring, and enabling-condition failures.
RLS	Compares residual welfare among survivors	Supplies dimension assignments, profiles, pathways, dependence tags, and uncertainty records.

2.2 Normative status

This document is a companion measurement standard. Its classification and record rules are normative for any implementation claiming WDBIP v1.4 conformance. Its proposed instruments, validation studies, example indicators, and research thresholds are informative until separately validated or adopted through a governed profile.

2.3 Non-claims

WDBIP does not claim:

- that welfare naturally consists of exactly seven metaphysically exhaustive dimensions;
- that every dimension has achieved cross-cultural measurement validity;
- that a survey response proves a metaphysical, spiritual, or causal claim;
- that correlation establishes causation;
- that every dimension must be active in every decision;
- that every relationship is beneficial or decision-material;
- that one scalar can represent the whole welfare profile;
- that RLS scores are objective truth;
- that internal consistency is empirical validation;
- that WDBIP creates authority, legal permission, or execution safety.

2.4 PCC attachment and source-of-truth rule

A WDBIP record is a PCC-linked measurement record, not a parallel decision authority. For every claim-bearing Tier 2 or Tier 3 use, the run's PCC SHALL record or reference at minimum:

- PCC.WDBIP.ProtocolVersion;
- PCC.WDBIP.SchemaVersion;
- PCC.WDBIP.RecordID;
- PCC.WDBIP.RecordHash or immutable artifact locator;
- PCC.WDBIP.RecordStatus;
- PCC.WDBIP.UnresolvedBoundaryItems;
- PCC.WDBIP.ScopeRedundancyMethod;
- PCC.WDBIP.GateRouting;
- reviewer identity and independence status where required.

The exact WDBIP packet supplies the detailed token, profile, pathway, dependence, and uncertainty records. The PCC supplies the run-level pointer and governing disclosure. If a PCC summary and the referenced WDBIP packet diverge, the discrepancy is a record-integrity failure and the stronger claim must be narrowed or refused until corrected.

2.5 WDBIP-SGP measurement and standing boundary

WDBIP classifies and measures welfare states, determinants, consequences, and interactions. SGP governs moral-patient evidence, protection posture, participation readiness, role readiness, intelligence profiling, and the controlled welfare-inclusion hypothesis interface. Therefore:

- a WDBIP score does not establish sentience, FPP, intelligence, authority, or personhood;
- SGP status does not substitute for evidence about the actual welfare state being scored;
- protection-relevant uncertainty routes through SGP and the Canon's precautionary interface;
- WDBIP records the measured welfare claim, while SGP records why and how the bearer enters protection and welfare-attention analysis;
- protection is not authority, and measurement is not standing.

Where SGP is decision-material, the WDBIP packet SHALL include the exact SGP record version or immutable reference.

3. Canonical welfare dimensions and boundary rules

3.1 Canonical definitions

Code	Dimension	Canonical focus	Includes	Does not automatically include
D1	Material	Resources, infrastructure, and subsistence security	Income, housing, food, energy, transport, productive assets, affordability, resource continuity	Health outcomes, social belonging, knowledge, autonomy, purpose, or ecosystem integrity merely because resources influence them
D2	Health	Physical and mental functioning, morbidity, injury, disability, and mortality risk	Bodily function, mental function, symptoms, pain, fatigue, sleep, vitality, recovery, safety-related health outcomes	Every social, material, spiritual, or environmental determinant of health

Code	Dimension	Canonical focus	Includes	Does not automatically include
D3	Social	Trust, belonging, relationships, care, and relational integrity	Support, cohesion, inclusion, reciprocity, conflict, violence, isolation, relational safety	Individual purpose, formal rights, knowledge quality, or resource adequacy unless directly represented as a social condition
D4	Knowledge	Epistemic access, learning, understanding, and correction capacity	Education, information access, accuracy, skills, competence, monitoring, error correction	Mere persuasion, authority, preference, or unverified confidence
D5	Agency	Autonomy, effective choice, participation, and freedom from coercion	Consent, real options, mobility, voice, appeal, control, non-domination	Satisfaction, competence alone, nominal choice, or authority merely inferred from intelligence
D6	Meaning	Purpose, coherence, mattering, identity, values, and valued projects	Existential coherence, purpose, value congruence, continuity, spiritual or secular connection, moral injury	Metaphysical truth, divine status, clinical health merely because meaning affects health, or propaganda presented as shared purpose
D7	Environment	Ecological and built-context integrity	Air, water, soil, climate exposure, habitat, biodiversity, pollution, regeneration, built-environment condition	The resulting human disease burden, material cost, or social conflict unless separately recorded in those dimensions

3.2 One-token one-primary-dimension rule

For a declared bearer, affected population, baseline, time window, and option, each material effect token SHALL have one primary welfare dimension. Scope assignment is governed separately by the Canon's Union Scope and redundancy rules.

An effect token is a bounded claim such as:

- "annual average PM2.5 exposure decreases by 4 micrograms per cubic meter";
- "self-reported loneliness decreases";
- "household rent burden increases by 5 percentage points";
- "employees gain a binding appeal right";
- "reported purpose in work improves".

A token MAY produce further effects in other dimensions, but those consequences must be separately defined and linked by a pathway record.

3.2A Canonical Union Scope binding

WDBIP does not define or own Union Scope redundancy methods. Each effect token SHALL declare its scope treatment using the methods owned by the pinned RippleLogic Canon: EMERGENT_SCALE_JUSTIFICATION, DEDUPLICATION, or ALLOCATION. The Canon's effect-token definition, instance-to-scope mapping, reduced-scope rule, and per-token allocation constraint control. WDBIP adds the obligation to record the selected treatment, its rationale, and its PCC reference.

Multi-scope stakeholder mapping is permitted. Silent multiplication of the same underlying welfare effect across scopes is not. Allocation must never suppress rights-covered, catastrophe-relevant, or CSV-material harm.

3.2B Scope assignment record

Each material token SHALL record:

- all represented Union Scopes;
- redundancy-handling method;
- scope rationale;
- primary scope when DEDUPLICATION is used;
- distinct scale-specific token identifiers and evidence when EMERGENT_SCALE_JUSTIFICATION is used;
- allocation coefficients where ALLOCATION is used;
- the allocation remainder and its rationale when the coefficients sum to less than 1;
- PCC Scope Coverage Declaration reference where reduced-scope mode applies.

A distinct downstream consequence at another scope requires a distinct target token and a token-level pathway. A repeated label at another scope is not automatically a distinct consequence. EMERGENT_SCALE_JUSTIFICATION does not license one vague token to be copied across scales; the scale-specific states must be separately defined.

3.2C Allocate-versus-split decision procedure

Use ALLOCATION only when one conserved welfare state or effect is being represented across several analytical scopes. Create distinct tokens when the bearers, states, baselines, time windows, mechanisms, or evidence objects differ materially.

Apply the following sequence:

1. **Same bearer and state?** If not, split into distinct tokens.
2. **Same causal event and measurement object?** If not, split.
3. **Conserved quantity or burden?** If yes, allocation may be appropriate. If each scope has its own realized state, split.
4. **Would separate entries create repeated credit or harm for one state?** If yes, deduplicate or allocate.
5. **Would allocation hide a distinct subgroup, rights, ruin, or structural effect?** If yes, split and route the gate-material issue.
6. **Do coefficients sum below 1?** Record the residual $1 - \text{sum}(\alpha_u)$ and classify it as justified attenuation, out-of-boundary remainder, measurement loss, or unresolved allocation. An unexplained residual is not conformant.

Worked contrast:

Situation	Correct treatment	Reason
One student's verified respiratory improvement appears relevant to Self, Household, Community, and Polity	DEDUPLICATION or bounded ALLOCATION	One bearer and one conserved health state must not be multiplied.
City maintenance expenditure and household travel-cost change arise from the same project	Distinct D1 tokens	Different bearers, states, evidence, and incidence.

Situation	Correct treatment	Reason
Loss of one sacred wetland creates ecological degradation, community displacement, and cultural-identity loss	Distinct D7, D3, and D6 tokens linked by pathways	The outcomes are causally related but are not the same state.

3.3 Primary-location test

Assign the primary dimension by asking, in order:

1. What is the immediate object or state being changed?
2. Who or what bears that state?
3. Is the claim a resource, health state, relationship, epistemic condition, effective freedom, meaning state, or environmental condition?
4. Would the claimed state still exist conceptually if the downstream consequence did not occur?
5. Is the same effect already represented elsewhere under another label?
6. Does the evidence directly measure the state or only a determinant, proxy, or consequence?
7. Is the effect actually rights-, catastrophe-, or CSV-material rather than merely residual welfare?

If the answer remains ambiguous, mark BOUNDARY_CONTESTED, preserve competing assignments in sensitivity analysis, and narrow any strong ranking claim.

3.3A Boundary contestation and classification reliability

BOUNDARY_CONTESTED is a legitimate uncertainty state, not a failure to be hidden. However, systematic contestation may indicate that definitions, anchors, training, or the dimension partition require revision.

A validation study SHALL preregister:

- the unit of coding;
- rater training and independence;
- the agreement statistic appropriate to nominal or ordinal labels;
- confidence intervals;
- the permitted adjudication process;
- the review trigger for excessive contestation.

The following are **provisional review triggers, not universal validity constants**:

Surface	Development target	High-stakes target	Review trigger
Primary-dimension assignment	agreement coefficient >= 0.70	>= 0.80	lower confidence bound below 0.60 or persistent domain-specific disagreement
Pathway-type assignment	>= 0.70	>= 0.80	materially lower agreement after training
BOUNDARY_CONTESTED rate	ordinarily < 20%	ordinarily < 10% for decision-critical tokens	exceedance requires category review, additional anchors, or narrower claims

Suitable statistics may include Krippendorff's alpha, Gwet's AC1/AC2, weighted kappa, or an intraclass coefficient where the scale and design warrant it. The statistic must be selected before outcomes are observed. Prevalence-sensitive statistics should be interpreted with their known limitations. Meeting a trigger does not establish construct validity; failing it requires review.

3.4 State, determinant, and consequence rule

WDBIP distinguishes:

- **State:** the condition directly represented in a dimension.
- **Driver or exposure:** a factor that may alter a state.

- **Mediator:** an intermediate mechanism.
- **Outcome:** a later state affected by the pathway.
- **Moderator:** a condition changing pathway strength or direction.
- **Enabling condition:** a condition without which the state cannot be maintained.
- **Indicator:** a measurement object, not necessarily the state itself.

Example:

Object	Role	Dimension
Air pollution concentration	Environmental state/exposure	D7
Inflammatory response	Mediator	D2
Asthma exacerbation	Health outcome	D2
Missed work and lost income	Material consequence	D1
Family caregiving strain	Social consequence	D3

4. Welfare Dimension Profile

4.1 Profile-before-index rule

A dimension SHALL be represented as a profile before a scalar is used. The minimum profile for Union Scope *u*, dimension *d*, and time *t* is:

W(u,d,t) = <S, F, C, X, L, G, U>

where:

Field	Meaning
S	Observed or estimated state
F	Functioning or realized performance
C	Capacity, resilience, and recovery potential
X	Exposures, drivers, dependencies, and enabling conditions
L	Lived or reported experience where applicable
G	Distribution across affected groups and worst-affected subgroups
U	Uncertainty, evidence status, staleness, and disagreement

The tuple is a record structure, not a formula requiring arithmetic combination.

4.1A Tuple-to-record crosswalk

The seven tuple fields organize the longer record required by Section 4.2:

Tuple field	Section 4.2 content primarily assigned here
S	direct state indicators, baseline, measurement date, option-relative change
F	functioning and realized-performance indicators
C	resilience, recovery, adaptive capacity, continuity potential

Tuple field	Section 4.2 content primarily assigned here
X	drivers, exposures, dependencies, enabling conditions, causal context
L	lived experience and affected-party evidence where appropriate
G	population distribution, incidence, worst-affected subgroup, severity concentration
U	evidence status, instrument version, provenance, missingness, staleness, confidence, disagreement, sensitivity

A field may reference another record rather than duplicate it, but the reference must be immutable and replayable.

4.2 Profile fields

Every decision-material profile SHOULD record:

- profile identifier and version;
- option, baseline, scope, population, and subgroup;
- dimension code;
- time horizon and measurement date;
- direct state indicators;
- functioning indicators;
- resilience or recovery indicators;
- drivers and enabling conditions;
- lived-experience indicators, where ethically and methodologically appropriate;
- distribution and worst-affected subgroup;
- measurement reference and instrument version;
- data source and provenance;
- missingness and staleness;
- uncertainty interval or qualitative class;
- reviewer confidence;
- claim action: ALLOW, NARROW, ESCALATE, REFUSE, MONITOR_ONLY, or SENSITIVITY_ONLY.

4.3 No scalar laundering

Aggregation may legitimately improve statistical reliability when several partially independent indicators measure the same construct. It does not automatically improve construct validity, causal validity, population transportability, or evidential maturity. A composite index SHALL NOT claim stronger validity or causal warrant than its weakest decision-material evidence dependency unless independent validation supports that stronger claim. A scalar MAY be used only when:

- the construct and intended use are defined;
- component direction and aggregation rules are declared;
- missing-data treatment is declared;
- reliability gains from aggregation are reported separately from validity and causal-warrant claims;
- reliability and validity evidence are appropriate to the population and context;
- distributional harms are not hidden;
- sensitivity to alternative plausible constructions is reported;
- the number is not used to bypass RF/NCRC, TRC, or CSV.

4.4 Weight legitimacy and sensitivity boundary

WDBIP protects the integrity of evidence entering each dimension. It does not establish the legitimacy of the Canon's cross-scope or cross-dimensional trade-off weights and does not redefine the RLS equation.

For every claim-bearing run in which a plausible alternative weight set changes the selected option, decisiveness status, or a material subgroup conclusion, the PCC SHALL record WEIGHT_SENSITIVE, the weight families tested, and the authority path used. A WDBIP-conformant record must not imply that careful measurement removes the normative and political character of weight selection.

At minimum, Tier 3 analysis SHALL test:

- constitutional floor-preserving alternatives;
- plausible stakeholder or participatory weight sets;
- a transparent equal-weight or reference-weight comparator where admissible;
- the effect of excluding any empirically unsupported local prominence adjustment;
- rank and decisiveness stability rather than score variation alone.

Weight sensitivity cannot rescue an option that fails RF/NCRC, TRC, or CSV.

5. Token-level cross-dimensional and cross-scope interaction records

5.1 Token-level interaction record

A pathway is defined between bounded effect tokens, not merely between dimension labels. For source token *s* and target token *t*:

I(s -> t) = <M, D, tau, beta, E, U>

The dimension pair dim(s) -> dim(t) and scope pair scope(s) -> scope(t) are derived labels. Two different mechanisms between the same dimension pair require separate pathway records. A dimension-level arrow is a summary view, not the primary evidentiary object.

where:

Symbol	Meaning
M	Proposed mechanism and pathway type
D	Direction: beneficial, harmful, mixed, nonlinear, or unknown
tau	Time lag or temporal profile
beta	Estimated magnitude or bounded qualitative effect
E	Evidence source, design, and dependence status
U	Uncertainty, alternative explanations, and sensitivity relevance

The record MUST also identify source token, target token, source and target scope, scope relation, population, baseline, option, and version.

5.2 Machine-token namespace rule

Human-readable labels may remain concise, but machine-readable values are field-namespaced. A pathway field SHALL use PATHWAY_* tokens, an evidence field SHALL use EVIDENCE_* tokens, and a direction field SHALL use DIRECTION_* tokens. A token from one namespace is invalid in another field.

5.3 Pathway types

Allowed pathway types are:

Token	Meaning
PATHWAY_CAUSAL_DIRECT	Evidence supports a direct causal effect within the declared model.
PATHWAY_CAUSAL_MEDIATED	Effect operates through one or more declared mediators.

Token	Meaning
PATHWAY_ENABLING_CONDITION	Source condition enables or constrains the target state.
PATHWAY_MODERATOR	Source changes the magnitude or direction of another pathway.
PATHWAY_FEEDBACK	Source and target influence one another over time.
PATHWAY_SHARED_CAUSE	Association may arise from a common cause rather than direct influence.
PATHWAY_CORRELATIONAL_ONLY	Association is observed but causal direction is not established.
PATHWAY_HYPOTHESIZED	Plausible mechanism exists but pathway evidence is insufficient.
PATHWAY_CONTESTED	Credible evidence or interpretation conflicts about the pathway.
PATHWAY_UNKNOWN	Direction or mechanism is not established.
PATHWAY_NOT_MATERIAL	Pathway is below the declared materiality boundary, with rationale and reopen trigger.

5.4 Causal language discipline

Use "causes," "produces," or "leads to" only when the study design and assumptions support that inference. Otherwise use:

- associated with;
- may affect;
- is a plausible pathway to;
- predicts under the declared model;
- is consistent with;
- remains unresolved.

A directed arrow in a diagram is not causal evidence.

5.5 Feedback and dynamic systems

Bidirectional relationships SHALL be represented as two pathways or a declared feedback structure. For example:

D6 meaning distress -> D2 mental-health impairment

and

D2 prolonged illness -> D6 disruption of valued projects

The two directions may have different mechanisms, evidence, magnitudes, and time lags.

6. Double-counting and dependence control

6.1 Core prohibition

The same underlying effect SHALL NOT receive multiple welfare contributions merely because it can be described in several ways.

6.2 Duplicate-effect test

Two proposed effect tokens are presumptive duplicates when they share all or most of:

- bearer;
- state changed;
- time window;

- baseline;
- causal event;
- instrument items;
- data source;
- mechanism;
- outcome.

Presumptive duplicates must be merged, assigned a primary location, or modeled with declared dependence rather than treated as independent gains or harms.

6.3 Distinct-consequence permission

One source event may legitimately create several distinct effects.

Example: closing a rural clinic may produce:

- D1: higher travel and out-of-pocket costs;
- D2: delayed diagnosis and health deterioration;
- D5: reduced effective access and choice;
- D3: caregiver burden and relationship strain.

These are not duplicates if each effect has a distinct state, bearer, evidence record, and pathway. They remain dependent and must not be represented as statistically independent without justification.

6.4 Evidence-dependence graph

Decision-material evidence objects SHOULD be represented in an evidence-dependence graph. At minimum, declare:

- shared source data;
- shared survey items;
- overlapping samples;
- repeated outcome measures;
- common mediators;
- common model assumptions;
- derived variables;
- institutional or commercial conflicts;
- duplicated qualitative testimony.

WDBIP dependence tags may inform the Canon's $\mathrm{kappa}(u,d)$ or other governed dependence controls. WDBIP does not redefine the canonical RLS equation.

6.5 No proxy multiplication

Several proxies for one latent construct do not automatically create several benefits. For example, purpose, life direction, and existential coherence may support a D6 profile, but they are not three independent welfare gains merely because three questionnaire items exist.

6.5A Construct-separation methods

Where several indicators may reflect one general factor, overlapping specific factors, mutually reinforcing symptom or state networks, or genuinely distinct constructs, the analyst SHALL not infer the structure from item labels alone. Depending on the research question and data, admissible methods may include:

- confirmatory factor analysis;
- exploratory structural equation modeling;
- bifactor or bifactor-ESEM models;
- network psychometrics;
- multitrait-multimethod analysis;
- item-response or differential-item-function analysis;
- qualitative cognitive interviewing and content-validity review.

No single method is universally controlling. Model choice, identification assumptions, fit criteria, cross-validation, alternative models, and interpretability must be preregistered or transparently reported. Better statistical fit alone does not prove the ontology of a welfare dimension.

6.6 Cross-scope duplicate-effect test

When the same token or conserved causal unit appears in more than one Union Scope, the record SHALL apply the Canon's declared redundancy treatment. Reviewers must ask:

1. Is the state materially different at each scale, or merely relabeled?
2. Are the bearers, mechanisms, time horizons, and evidence objects distinct?
3. Has EMERGENT_SCALE_JUSTIFICATION, DEDUPLICATION, or ALLOCATION been declared?
4. If allocation is used, do coefficients sum to no more than 1 for the conserved token?
5. Does any treatment weaken a rights, catastrophe, or CSV signal?

A failure of this test requires merge, allocation repair, sensitivity analysis, or claim narrowing before RLS is recomputed.

7. Health boundary protocol

7.1 D2 Health definition

D2 Health records physical and mental states, functioning, morbidity, injury, disability, recovery, and mortality risk. It may include:

- bodily integrity and physiological functioning;
- disease and symptom burden;
- mental and emotional functioning;
- pain, fatigue, sleep, and vitality;
- developmental functioning;
- functional capacity and disability;
- resilience and recovery;
- health-related safety and mortality risk;
- lived health experience where appropriately measured.

7.2 Determinants are not identical to health

Material, social, epistemic, agency, meaning, and environmental conditions may determine health. They retain their primary dimensions when the immediate state is not itself health.

Claim	Primary location	Linked pathway
Housing is unaffordable	D1 Material	D1 -> D2 if stress or illness effects are evidenced
A person is socially isolated	D3 Social	D3 -> D2 if health effects are evidenced
A worker lacks schedule control	D5 Agency	D5 -> D2 if health effects are evidenced
A person experiences existential despair	D6 Meaning	D6 -> D2 if clinical or functional health consequences are evidenced
Air contains harmful particulate matter	D7 Environment	D7 -> D2 if exposure and health effects are evidenced

7.3 Physical and mental health are not exhaustive flourishing

Health is necessary and often foundational, but it does not subsume material sufficiency, relationship quality, knowledge, autonomy, meaning, or ecological integrity. A person can be clinically healthy while lacking freedom, purpose, secure housing, education, or a viable environment.

8. Meaning, spiritual experience, and metaphysical claims

8.1 D6 Meaning definition

D6 Meaning records purpose, coherence, mattering, identity, value congruence, continuity of valued projects, and connection beyond the isolated self.

8.2 Meaning Integrity Profile

A D6 profile MAY include:

Component	Operational question	Example indicators
Coherence	Is life or collective experience sufficiently comprehensible and integrated?	Narrative continuity, comprehensibility, identity integration, existential disorientation
Purpose	Are valued future directions present and actionable?	Goal direction, future orientation, persistence, purpose presence
Mattering	Does the subject experience significance and contribution?	Perceived significance, burdensomeness, recognition, contribution
Value congruence	Are actions and institutions aligned with declared values?	Moral injury, value-action conflict, ethical compromise
Continuity	Can valued identities, cultures, relationships, and projects continue?	Cultural continuity, intergenerational transmission, project continuity
Connection or transcendence	Is there experienced connection beyond the isolated self?	Nature connection, service, spirituality, religion, humanity, AIU, art, truth
Existential distress	Is meaning disruption causing material suffering or dysfunction?	Hopelessness, nihilistic distress, spiritual injury, identity collapse

8.3 Spiritual-measurement boundary

WDBIP permits measurement of reported and observable aspects of spiritual or existential well-being, including:

- inner peace;
- spiritual struggle or distress;
- hope;
- purpose;
- value congruence;
- experienced transcendence;
- connection to community, nature, humanity, or ultimate reality;
- freedom to hold, reject, change, and practice beliefs;
- functional and health consequences.

WDBIP does not permit an RLS score to establish:

- that God exists or does not exist;
- that AIU is empirically proven;
- that a person is enlightened;
- that one religion is metaphysically correct;
- that a reported spiritual experience verifies its external interpretation;
- that religious conformity is welfare.

The measured claim is the experience, freedom, functioning, relationship, or consequence, not the metaphysical truth of the belief.

8.4 Primary locations for spiritual and existential effects

Effect	Primary location
Experienced purpose, coherence, spiritual peace, or distress	D6 Meaning
Depression, anxiety, sleep loss, or functional impairment associated with spiritual distress	D2 Health
Spiritual community, belonging, support, exclusion, or conflict	D3 Social
Freedom of belief, conscience, worship, exit, or nonbelief	D5 Agency and potentially RF/NCRC
Access to material resources or places necessary for practice	D1 Material or D7 Environment, depending on the state
Knowledge about a tradition or worldview	D4 Knowledge
Claimed supernatural or metaphysical fact	RG claim-status record, not a welfare score

8.5 Anti-coercion rule

No institution may improve a D6 score by forcing belief, suppressing dissent, mandating positive reports, or treating conformity as meaning. Coercion routes to D5 Agency and RF/NCRC review.

9. Environmental boundary protocol

9.1 D7 Environment definition

D7 Environment records ecological and built-context integrity, including:

- air and water quality;
- soil condition;
- habitat and biodiversity;
- climate stability and exposure;
- pollution and toxic burden;
- ecosystem functioning and regeneration;
- resource cycles;
- built-environment safety and accessibility;
- resilience of environmental enabling systems.

9.2 Environmental health distinction

"Environmental health" has two meanings that SHALL be separated:

1. **Condition of the environment:** D7.
2. **Health consequence of environmental exposure:** D2.

Example:

D7 degraded air quality -> D2 respiratory morbidity

The D7 record measures the environmental state. The D2 record measures the health outcome. Cost, social, knowledge, agency, and meaning consequences are separately recorded when material.

9.3 Non-sentient enabling value

A non-sentient ecosystem may have major relevance because it enables sentient welfare, future options, resilience, cultural continuity, and structural viability. WDBIP does not infer sentience from ecological importance and does not require direct phenomenal welfare for D7 to be active.

9.4 Welfare measurement versus moral standing

D7 enabling value, a D2 health state, or any other WDBIP measurement does not itself determine moral standing. Where a bearer may be a moral patient, SGP governs the evidence and protection posture. Where the bearer is a non-sentient system, WDBIP may still record ecological, structural, cultural, option, or enabling value without asserting phenomenal welfare. The two records answer different questions:

- **WDBIP:** What state changes, for whom or what, with what evidence and interaction structure?
- **SGP:** What protection and welfare-attention posture is warranted for the bearer under uncertainty?

10. Dimension-specific minimum profiles

10.1 D1 Material

Minimum profile domains:

- income and wealth;
- food, housing, energy, water, and transport security;
- affordability and burden;
- infrastructure access and reliability;
- productive capacity and livelihood continuity;
- distribution, deprivation, and resilience to shock.

Boundary warning: material resources may enable agency, health, knowledge, social participation, and meaning, but those downstream states require distinct evidence.

10.2 D2 Health

Minimum profile domains:

- morbidity and mortality;
- physical and mental functioning;
- symptoms, pain, fatigue, sleep, and vitality;
- disability and participation limitation;
- safety-related health risk;
- care access, recovery, and resilience;
- subgroup inequalities.

Boundary warning: determinants of health retain their own dimensions.

10.3 D3 Social

Minimum profile domains:

- trust and belonging;
- supportive relationships and care;
- reciprocity and inclusion;
- loneliness and isolation;
- conflict, violence, discrimination, and relational safety;
- institutional and community cohesion;
- distribution of voice and recognition.

Boundary warning: social approval is not authority, truth, meaning, or rights compliance.

10.4 D4 Knowledge

Minimum profile domains:

- access to information and education;
- accuracy, completeness, provenance, and uncertainty;
- learning and skill development;
- epistemic diversity and contestability;
- monitoring, feedback, and correction capacity;
- misinformation, secrecy, and knowledge loss;
- distribution of access and competence.

Boundary warning: confidence, credentials, or volume of information are not automatically knowledge quality.

10.5 D5 Agency

Minimum profile domains:

- real options and freedom of action;
- consent and freedom from coercion;

- participation, representation, and appeal;
- control over consequential conditions;
- mobility, exit, and remedy;
- dependency and domination risk;
- distribution of effective power.

Boundary warning: nominal choice, compliance, intelligence, or participation theater is not effective agency.

10.6 D6 Meaning

Minimum profile domains:

- purpose;
- coherence;
- mattering;
- identity and continuity;
- value congruence and moral injury;
- connection or transcendence;
- existential distress;
- cultural and worldview pluralism.

Boundary warning: meaning is culturally and personally variable. It must not be imposed by the evaluator.

10.7 D7 Environment

Minimum profile domains:

- air, water, soil, and chemical condition;
- climate and disaster exposure;
- habitat, biodiversity, and ecosystem function;
- pollution and waste;
- regeneration and carrying capacity;
- built-environment quality;
- environmental distribution and injustice;
- monitoring and uncertainty.

Boundary warning: ecosystem condition and human health consequence are linked but distinct.

11. Interaction matrix

The following matrix identifies common pathway questions. It is a discovery aid, not a claim that every pathway is active.

Source -> Target	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
D1 Material	Resource feedback, affordability, asset compounding	Nutrition, housing, care access, stress	Participation costs, household strain	Education and information access	Real options, dependency, exit capacity	Security for valued projects or material alienation	Consumption, extraction, infrastructure burden
D2 Health	Care costs, earning capacity	Comorbidity, recovery, symptom networks	Caregiving, isolation, relationship strain	Cognitive function, school/work learning	Functional choice, consent capacity	Illness meaning, identity, valued-project continuity	Health-related environmental behavior and capacity
D3 Social	Mutual aid, employment networks, shared resources	Support, loneliness, violence, stress pathways	Trust feedback, conflict cycles	Knowledge sharing, misinformation networks	Participation, collective efficacy, domination	Belonging, shared identity, cultural continuity	Stewardship norms, collective-resource governance

Source -> Target	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
D4 Knowledge	Productivity, resource allocation, fraud reduction	Health literacy, clinical knowledge, risk communication	Shared understanding, polarization, trust	Learning feedback and correction	Informed consent, effective choice	Worldview coherence, education and purpose	Environmental monitoring, ecological knowledge
D5 Agency	Employment choice, access, distribution	Control, stress, care decisions	Voice, consent, conflict resolution	Inquiry freedom, access, censorship	Power feedback, institutional capture	Value congruence, authorship of life projects	Participation in environmental decisions and stewardship
D6 Meaning	Consumption priorities, work motivation, sacrifice	Hope, distress, coping, behavioral pathways	Belonging, solidarity, exclusion	Curiosity, interpretation, ideology	Purposeful action, commitment, coercive ideology	Identity feedback, purpose renewal, spiritual struggle	Nature connection, sacred places, stewardship orientation
D7 Environment	Resource availability, infrastructure loss, livelihood	Pollution, climate, disease, injury	Displacement, conflict, community place	Observation, monitoring, traditional ecological knowledge	Access, environmental justice, adaptive options	Place attachment, cultural loss, intergenerational purpose	Ecosystem feedback, tipping, regeneration

Every active pathway requires an interaction record. Empty or unsupported cells remain unknown or not material rather than being presumed active.

12. Evidence and uncertainty protocol

12.1 Evidence-status vocabulary

Status	Meaning
EVIDENCE_DIRECT_OBSERVATION	State is directly measured within declared limits.
EVIDENCE_VALIDATED_INSTRUMENT	Instrument has relevant reliability and validity evidence for the population and use.
EVIDENCE_EMPIRICALLY_ESTIMATED	Effect is estimated using a declared empirical design and model.
EVIDENCE_EXPERT_JUDGMENT	Structured expert judgment is used with rationale and uncertainty.
EVIDENCE_PARTICIPATORY	Affected-party testimony or deliberation contributes to the record.
EVIDENCE_QUALITATIVE_CODED	Qualitative evidence is coded using a declared protocol.
EVIDENCE_HYPOTHESIZED	Plausible but not adequately established.
EVIDENCE_CONTESTED	Credible disagreement exists.
EVIDENCE_UNKNOWN	Evidence is insufficient or absent.

Status	Meaning
EVIDENCE_NOT_MATERIAL	Below declared materiality threshold with rationale and reopen trigger.

12.2 Unknown is not zero

Missing evidence SHALL NOT be represented as no effect. Use EVIDENCE_UNKNOWN, a conservative bound where justified, claim narrowing, evidence collection, sensitivity analysis, or refusal of the stronger claim.

12.3 False precision

The numerical resolution of a score SHALL NOT exceed the evidence resolution. A qualitative testimony-coded estimate should not be presented with several decimal places unless the decimals arise from a governed transformation whose uncertainty remains visible.

12.4 Cross-cultural and cross-substrate measurement

A measure used across cultures, languages, groups, ages, or substrates requires evidence that the construct and items retain sufficiently comparable meaning for the intended claim.

For latent-variable instruments, the normal test sequence is:

1. **Configural invariance:** sufficiently comparable factor pattern or construct organization.
2. **Metric invariance:** sufficiently comparable loadings, supporting comparison of associations and relationships.
3. **Scalar invariance:** sufficiently comparable intercepts or thresholds, normally required for direct latent-mean comparison.
4. **Residual or strict invariance:** additional equality of residuals where the intended claim requires it.

Justified partial invariance or approximate-invariance methods may be used when preregistered and transparently bounded. Direct group ranking or mean comparison requires scalar or defensible partial-scalar support; configural similarity alone is insufficient.

Where measurement invariance is not established:

- report group-specific interpretations;
- avoid direct mean comparisons;
- use qualitative or mixed-method evidence;
- conduct differential-item-function review;
- narrow the claim;
- retain uncertainty.

12.5 Affected-party evidence and research-ethics routing

Lived experience is evidence about experienced states, barriers, meaning, relationships, and consequences. It is not automatically a complete causal explanation. High-stakes evaluation SHOULD combine affected-party evidence with relevant behavioral, administrative, environmental, clinical, structural, and longitudinal evidence.

Measurement design must not create the harm it purports to measure. Coercive participation, unsafe disclosure, manipulative elicitation, unnecessary collection of sensitive beliefs, or disproportionate burden on vulnerable groups reopens RF/NCRC. Material privacy, security, staffing, monitoring, or implementation failures reopen CSV. A waiver or consent form does not erase domination, rights, or structural concerns.

12.6 Time-window coherence

Every token and pathway SHALL declare its measurement, exposure, and outcome windows. Tokens may be compared in one run only when the windows are aligned or when a declared temporal model explains the mismatch.

A short-term benefit must not be compared against a long-term harm as though both were contemporaneous and equally observed. Tier 2 and Tier 3 records SHALL identify:

- baseline period;
- observation or forecast period;
- lag structure;
- duration and reversibility;

- discounting or temporal weighting source where used;
- monitoring points and reopen triggers.

A material window mismatch without a causal or decision rationale requires narrowing, sensitivity analysis, or refusal of the stronger comparison.

12.7 Severe subgroup-harm visibility

The distribution field G is not satisfied by reporting an average alone. For every materially affected population, the record SHALL search for and report worst-affected subgroups consistent with the Canon's subgroup rules.

A favorable aggregate cannot hide a severe concentrated harm. Where a subgroup effect may be rights-covered, catastrophic, irreversible, domination-relevant, or structurally disabling, it routes to the relevant gate rather than remaining an offsettable RLS contribution.

For residual welfare effects, the record SHALL disclose whether the aggregate sign differs from the worst-affected subgroup sign. If a small subgroup bears a large harm, the record must preserve the harm token, uncertainty, incidence, and sensitivity; it may not be averaged away or omitted because the population mean is positive.

13. Tier-proportional implementation

Tier	Minimum WDBIP obligation
Tier 1	Identify material dimensions, assign each effect token one primary location, note obvious cross-dimensional pathways, and confirm no duplicate counting. A brief qualitative record is sufficient.
Tier 2	Complete profiles for material cells; record source-target pathways, evidence status, time lag, subgroup distribution, uncertainty, and dependence; conduct boundary and sensitivity review for contested effects.
Tier 3	Preregister constructs, instruments, baselines, causal graph, interaction hypotheses, missing-data rules, subgroup discovery, measurement-invariance requirements, dependence treatment, independent review, and revision triggers. Publish or preserve the replay packet.

A lower tier must not be selected to avoid rights, catastrophe, structural, physical, environmental, subgroup, or cultural-measurement safeguards.

14. Record status and routing

14.1 Record statuses

WDBIP record statuses describe measurement completeness only:

- WDBIP_RECORD_COMPLETE
- WDBIP_RECORD_COMPLETE_WITH_UNCERTAINTY
- WDBIP_RECORD_INCOMPLETE
- WDBIP_RECORD_NOT_MATERIAL

They are not pass/fail verdicts for an option.

14.2 Routing rules

Finding	Required routing
Unsupported or ambiguous dimension boundary	Narrow, sensitivity-only, collect evidence, or escalate through RG/category grounding.
Duplicate or dependent effects	Merge, model dependence, adjust governed dependence controls, and recompute RLS if necessary.
New severe rights evidence	Reopen RF/NCRC.
New catastrophe, irreversible, or ruin pathway	Reopen TRC.
New dependency, capacity, containment, environmental, or implementation failure	Reopen CSV and TRC if ruin-relevant.
Instrument invalid for population or culture	Narrow claim, replace instrument, collect evidence, or refuse comparison.
Meaning or spiritual claim exceeds measurable experience	Separate RG metaphysical claim from D6 experiential evidence.
No material welfare interaction	Record EVIDENCE_NOT_MATERIAL with rationale and reopen trigger.

15. Validation, falsification, and dimension-set migration

15.1 Coverage hypothesis

The seven-dimensional field detects materially relevant benefits, harms, dependencies, and affected groups that simpler comparator models omit.

Weakening evidence: no additional material detection, high false-positive burden, or equal performance from a simpler model.

15.2 Discriminant-validity hypothesis

The seven dimensions remain meaningfully distinguishable in theory, evidence coding, and rater application.

Weakening evidence: persistent inability to distinguish dimensions, very high correlations across most domains without distinct mechanisms, or frequent arbitrary reassignment.

15.3 Incremental-value hypothesis

Each dimension adds decision-relevant information beyond the other six. Decision value is not assumed to be one universal scalar. Before testing, preregister one or more metrics *V_k* from the outcome families in Section 15.9. For dimension *d* and metric *k*:

Delta V(d,k) = V_k(seven-dimension model) - V_k(model without d)

Admissible *V_k* measures include omission rate, rater reliability, predictive calibration, subgroup detection, realized adverse events, decision reversals, implementation burden, and affected-party assessment. Directionality, practical importance, and normalization must be declared before results are observed. If incremental value is repeatedly negligible or adverse across decision-material metrics, the dimension should be narrowed, merged, or redesigned.

15.4 Reliability hypothesis

Independent trained raters applying the same evidence reach acceptably similar primary-location, pathway-type, scope-treatment, and uncertainty classifications. The preregistration requirements and provisional review triggers in Section 3.3A apply.

Failure requires clearer definitions, anchors, examples, training, or revised categories.

15.5 Predictive-pathway hypothesis

Preregistered pathways predict later target outcomes beyond baseline values and plausible confounders. Out-of-sample performance, calibration, missingness, and distribution shift must be reported.

Failure weakens the specific pathway, not automatically the whole dimension architecture.

15.6 Intervention-sensitivity hypothesis

A targeted intervention should first change the construct it directly addresses and only then support claims about downstream ripples.

Examples:

- an air-quality intervention should change D7 exposure indicators before a D2 health benefit is claimed;
- a belonging intervention should change D3 indicators before D6 or D2 consequences are asserted;
- a purpose intervention should change D6 indicators before health benefits are generalized.

15.7 Cross-cultural and cross-substrate invariance hypothesis

Instruments intended for group comparison retain adequate construct meaning and measurement properties across those groups. Section 12.4 controls the level of comparison permitted by configural, metric, scalar, partial, or approximate invariance evidence.

Failure requires group-specific interpretation or withdrawal of direct comparative claims.

15.8 Anti-Goodhart hypothesis

WDBIP measures remain difficult to improve cosmetically while underlying welfare worsens.

Adversarial tests should include:

- forced positive-reporting;
- indicator substitution;
- survey coaching;
- exclusion of harmed subgroups;
- relabeling one effect across several dimensions;
- short-term gains masking long-term degradation;
- spiritual or cultural conformity presented as meaning;
- environmental offsets masking local harm;
- strategic scope allocation or unexplained allocation residuals;
- weight selection chosen after results are known.

15.9 Decision-utility hypothesis and named comparators

WDBIP improves consequential decisions enough to justify its evidence and audit burden.

A validation study SHOULD compare:

1. MathGov seven dimensions with WDBIP;
2. a health-only or narrow-outcome model;
3. the OECD Well-being Framework or a domain-appropriate dashboard implementation;
4. WHOQOL and, where appropriate, WHOQOL-SRPB;
5. a VanderWeele-style flourishing measure;
6. an Alkire-Foster-style multidimensional counting approach where deprivation identification is relevant;
7. ablation models removing one WDBIP dimension at a time.

The comparator set must match the decision domain. WDBIP must not claim superiority merely because another framework has a different purpose.

Outcomes should include omission rate, false-positive burden, rater reliability, predictive calibration, subgroup detection, realized adverse events, decision reversals, burden, affected-party assessment, weight sensitivity, and resistance to adversarial gaming.

15.10 Runnable validation study skeleton

The standalone WDBIP_VALIDATION_STUDY_PROTOCOL_v1.0.md is the controlling study-template companion for v1.4. At minimum, a preregistered validation study SHALL declare:

- decision domains and populations;

- sampling frame and power or precision analysis appropriate to each model;
- effect-token unit and annotation protocol;
- rater number, training, blinding, adjudication, and independence;
- named comparators and ablation models;
- primary and secondary metrics with directionality;
- reliability statistics and confidence intervals;
- construct and discriminant-validity models, including alternatives;
- configural, metric, scalar, partial, or approximate invariance plan;
- longitudinal and intervention tests where causal claims are attempted;
- missing-data, subgroup, time-window, and distribution-shift plans;
- weight-sensitivity and anti-Goodhart tests;
- revision, publication, and adverse-result rules.

Sample sizes shall be justified by simulation, precision, power analysis, or design-specific guidance rather than copied from one universal threshold. The protocol's numerical triggers are provisional governance triggers, not claims that the instrument is validated.

15.11 Dimension-set revision and migration rule

The governing taxonomy is versioned independently from the document. WDBIP v1.4 uses WELFARE_DIMENSION_SET_7D_V1.

If evidence supports narrowing, dividing, merging, adding, or removing a dimension:

1. issue a governed dimension-set proposal with construct definitions and evidence;
2. publish a crosswalk from the old taxonomy to the proposed taxonomy;
3. preserve all prior records and their original version pins;
4. prohibit silent retroactive recoding;
5. identify which historical comparisons remain valid, conditionally bridgeable, or non-comparable;
6. re-run active decisions when the change is material to ranking, subgroup harm, gate routing, or monitoring;
7. version schemas, validators, examples, and PCC pointers together;
8. preserve an appeal and challenge route.

A future eighth dimension is neither forbidden nor presumed. It requires a governed Canon revision and evidence that it improves coverage, discriminant validity, decision utility, or rights protection enough to justify additional complexity.

15.12 Partition-instability trigger

A domain-level category review is required when replicated studies show one or more of the following:

- persistent primary-location agreement below the preregistered target;
- BOUNDARY_CONTESTED rates above the preregistered review trigger;
- no stable discriminant structure across reasonable models;
- repeated cross-cultural non-invariance that defeats the intended use;
- negligible or adverse incremental decision value;
- systematic gaming or subgroup-harm masking;
- implementation burden disproportionate to decision value.

The review may refine definitions, improve anchors, permit structured multi-label discovery while retaining one scoring home, merge or split dimensions, or withdraw claims. It must not manipulate categories merely to preserve favorable scores.

16. Conformance requirements

A WDBIP-conformant material RLS record SHALL include:

1. exact Canon, WDBIP, schema, welfare-dimension-set, and relevant SGP version pins;
2. PCC record identifier and immutable WDBIP artifact reference;
3. option, baseline, declared scopes, population, and time boundary;
4. material effect-token registry;
5. one primary dimension for each token;
6. Canon-conformant represented scopes and redundancy treatment, including a primary scope only where DEDUPLICATION is used;
7. profiles for decision-material dimensions;

8. token-level interaction records for claimed cross-dimensional or cross-scope pathways;
9. field-namespaced pathway, evidence, direction, and boundary-status tokens;
10. evidence-status and uncertainty fields;
11. dependence, duplicate-effect, and cross-scope review;
12. subgroup distribution and worst-affected-group review;
13. cross-cultural or cross-group measurement boundary where relevant;
14. WDBIP record status;
15. routing of any gate-material findings;
16. monitoring and reopen triggers;
17. reviewer identity and independence status for high-stakes use.

17. Deterministic conformance vectors

Vector 1: Environmental exposure and health consequence

Input: Lower particulate concentration and fewer asthma attacks are entered as one combined Environment score and again as a Health score.

Expected: FAIL. Separate D7 environmental exposure token from D2 health-outcome token; link them through an evidenced pathway; disclose dependence.

Vector 2: Spiritual experience

Input: Respondents report increased inner peace after a voluntary contemplative program. Evaluator records "AIU proven" under Meaning.

Expected: FAIL. D6 may record reported inner peace or existential coherence. The metaphysical claim belongs in RG and is not established by the welfare report.

Vector 3: Job loss

Input: Job loss is scored as Material, Health, Social, Agency, and Meaning using the same administrative record without separate outcomes.

Expected: FAIL. Job and income loss are D1. Other dimensions require separate state definitions and evidence.

Vector 4: Distinct consequences

Input: Job loss is linked to verified income loss, clinically measured distress, loss of insurance choice, and household conflict.

Expected: PASS with dependence disclosure. The effects are distinct but share a common cause.

Vector 5: Correlation presented as causation

Input: Meaning and health scores correlate cross-sectionally. Record states that meaning causes health.

Expected: FAIL. Use `PATHWAY_CORRELATIONAL_ONLY` unless a design and assumptions support causal inference.

Vector 6: Unknown treated as neutral

Input: No data exist for community meaning. The cell is entered as zero.

Expected: FAIL. Use `EVIDENCE_UNKNOWN`, collect evidence, use sensitivity, or narrow the claim.

Vector 7: Eighth dimension

Input: "Spiritual Health" is added as D8 and scored alongside the canonical seven without a governed Canon revision.

Expected: FAIL. Measure spiritual-existential experience primarily within D6 and linked dimensions. WDBIP does not create D8.

Vector 8: Sixth gate

Input: Option passes all Canon gates but is rejected by a newly invented WDBIP pass/fail gate.

Expected: FAIL. WDBIP record status informs evidence quality and routing; it is not an option gate.

Vector 9: Gate-material discovery

Input: Welfare interaction analysis identifies an irreversible biosphere tipping pathway omitted from TRC.

Expected: PASS only if TRC is reopened before RLS.

Vector 10: Cultural non-invariance

Input: A purpose scale is used to rank national populations despite evidence that its items are not comparable across groups.

Expected: FAIL for direct comparative ranking. Use group-specific interpretation, validated alternative, mixed methods, or narrower claims.

Vector 11: Silent cross-scope multiplication

Input: The same one-time grant is scored in Household, Community, Organization, and Polity without a declared Canon redundancy method.

Expected: FAIL. Apply DEDUPLICATION, justified ALLOCATION, or EMERGENT_SCALE_JUSTIFICATION; attach the treatment to the PCC and recompute if necessary.

Vector 12: Measurement treated as moral standing

Input: A high D4 Knowledge score is used to infer FPP, governance authority, or low protection need.

Expected: FAIL. WDBIP measurement does not establish SGP standing, protection, personhood, or authority.

Vector 13: Namespace misuse

Input: PATHWAY_HYPOTHESIZED is entered in an evidence-status field.

Expected: FAIL. Evidence fields require EVIDENCE_* values; pathway fields require PATHWAY_* values.

Vector 14: Allocate versus split

Input: One city budget cost and one household travel-cost increase are combined as a single conserved token and allocated across U2 and U5.

Expected: FAIL. Different bearers and states require distinct D1 tokens. Allocation is for one conserved effect, not for causally related but distinct incidence.

Vector 15: Unexplained allocation residual

Input: A conserved token is allocated with coefficients summing to 0.65 and no explanation for the remaining 0.35.

Expected: FAIL. Record attenuation, out-of-boundary remainder, measurement loss, or unresolved allocation; otherwise the record is incomplete.

Vector 16: Time-window mismatch

Input: A one-month employment benefit is compared with a twenty-year pollution burden without a temporal model, weighting source, or sensitivity analysis.

Expected: FAIL. Align windows or disclose the causal and temporal comparison model; narrow the ranking until corrected.

Vector 17: Aggregate masks severe subgroup harm

Input: Average agency improves, but a small subgroup loses all meaningful exit and appeal. The subgroup harm is omitted because the mean is positive.

Expected: FAIL. Preserve the subgroup token and route potential domination or rights harm to RF/NCRC.

Vector 18: Weight-sensitive selection

Input: Two plausible floor-preserving weight sets select different options, but the record presents one unique WDBIP/RLS answer.

Expected: FAIL. Mark WEIGHT_SENSITIVE, disclose the sets and authority path, and do not misrepresent the result as a unique framework selection.

Vector 19: Dimension-set migration

Input: A future taxonomy merges D3 and D6, and an old record is silently relabeled without a crosswalk or rerun.

Expected: FAIL. Preserve the original record, issue a versioned mapping, state comparability, and rerun if material.

Vector 20: Excess boundary contestation

Input: Forty percent of decision-critical tokens remain BOUNDARY_CONTESTED after training, but the system claims validated classification.

Expected: FAIL for the validation claim. Trigger partition or anchor review and narrow use. The individual records may remain usable with disclosed uncertainty where no stronger claim depends on them.

18. Worked example: school green-corridor program

18.1 Decision

A city considers converting traffic lanes beside schools into shaded green corridors with trees, reduced vehicle access, walking paths, and community spaces.

18.2 Primary effect tokens

Token	Primary dimension	Scope treatment	Evidence posture
Tree-canopy coverage increases	D7 Environment	U3 Community	Direct observation / modeled projection
Roadside particulate exposure decreases	D7 Environment	U3 Community	Empirical estimate
Heat exposure during school travel decreases	D7 Environment	U3 Community	Modeled and monitored
Respiratory symptoms decrease	D2 Health	U1 Self; downstream household/community effects require separate tokens	Longitudinal health evidence required
Walking activity increases	D2 Health	U1 Self	Behavioral observation
Safe route choice increases	D5 Agency	U1 Self primary; DEDUPLICATION across U1/U2	Accessibility and choice evidence
Community use of shared space increases	D3 Social	U3 Community	Observation and affected-party evidence
Environmental learning opportunities increase	D4 Knowledge	U3 Community	Program and learning evidence
Individual nature connection increases	D6 Meaning	U1 Self	Validated or mixed-method evidence
Community place identity increases	D6 Meaning	U3 Community	Validated or mixed-method evidence
Construction and maintenance costs increase	D1 Material	U5 Polity	Budget evidence

18.3 Interaction records

Source -> target	Type	Claim boundary
D7 lower particulate exposure -> D2 respiratory health	PATHWAY_CAUSAL_MEDIATED	Requires exposure-response evidence and local monitoring.
D7 shade -> D2 heat-related health	PATHWAY_CAUSAL_DIRECT or PATHWAY_CAUSAL_MEDIATED	Depends on temperature, exposure, and health model.
D5 safe route choice -> D2 activity	PATHWAY_CAUSAL_MEDIATED	Requires behavior evidence; availability alone is not use.
D3 shared space -> D6 place meaning	PATHWAY_HYPOTHESIZED with EVIDENCE_EMPIRICALLY_ESTIMATED when supported	Do not assume community meaning from design rhetoric.
D4 ecological learning -> D5 stewardship agency	PATHWAY_CAUSAL_MEDIATED	Knowledge may enable action but does not guarantee it.
D1 project cost -> D5 access	PATHWAY_MODERATOR	Fees or displacement could limit access.

18.4 Double-counting and scope review

Do not count "cleaner air" as both a D7 environmental benefit and a D2 health benefit unless the D2 token is a separately defined health outcome. Do not count survey items for calm, connection to nature, and belonging as independent effects without checking whether they measure overlapping constructs.

The same student's respiratory improvement must not be multiplied across Self, Household, Community, and Polity merely because each scope has an interest in the outcome. Use one primary scope under DEDUPLICATION, or define separate scale-specific tokens and justify EMERGENT_SCALE_JUSTIFICATION. City expenditure and household travel-cost effects are distinct tokens with different bearers.

18.5 Gate feedback

If the project displaces disabled access, creates severe road-safety risk, or causes unjust property expropriation, RF/NCRC must be revisited. If tree species create material wildfire or infrastructure risk, CSV and possibly TRC must be revisited. WDBIP does not average those failures into residual benefits.

19. Compact implementation template

A. Run identity

- WDBIP version:
- Schema version:
- Welfare dimension-set version:
- Canon version:
- PCC record ID / WDBIP record hash:
- SGP record reference, if applicable:
- Option:
- Baseline:
- Declared Union Scopes:
- Scope Coverage Declaration reference, if applicable:
- Population and subgroups:
- Time horizon and comparison-window rule:
- Weight-sensitivity status / analysis reference:
- Boundary-reliability plan or status, where applicable:
- Dimension-migration reference, where applicable:
- Analyst and reviewer:

B. Effect-token registry

B1. Identity and classification

Token ID	Effect claim	Primary dimension	Bearer

B2. Scope treatment

Token ID	Represented scopes	Primary scope if deduplicated	Scope method

B3. Evidence boundary

Token ID	Baseline	Time	Evidence status	Uncertainty

C. Dimension profile

Field	Record
State	
Functioning	
Capacity/resilience	
Drivers/enabling conditions	
Lived experience	
Distribution/worst affected	
Uncertainty/evidence status	

D. Interaction registry

D1. Path identity and classification

Path ID	Source token	Target token	Pathway type	Scope relation

D2. Mechanism and effect

Path ID	Mechanism	Direction	Time lag	Magnitude

D3. Evidence and uncertainty

Path ID	Evidence	Alternatives	Uncertainty

E. Dependence review

- Shared data sources:
- Shared items or instruments:
- Common causes:
- Mediators:
- Duplicate tokens merged:
- Canonical dependence-control implications:
- Cross-scope redundancy method and rationale:
- Allocation coefficients, if used:
- Allocation residual and rationale, if sum < 1:
- Aggregate sign versus worst-affected-subgroup sign:

F. Routing

- RF/NCRC reopened?
- TRC reopened?
- CSV reopened?
- Claim narrowed?

- Sensitivity required?
- Monitoring/reopen trigger:

G. Record status

- WDBIP_RECORD_COMPLETE
- WDBIP_RECORD_COMPLETE_WITH_UNCERTAINTY
- WDBIP_RECORD_INCOMPLETE
- WDBIP_RECORD_NOT_MATERIAL

20. Diamond summary

The seven welfare dimensions are distinct lenses on one relational reality. WDBIP assigns each effect one primary dimensional home, binds its Union Scope treatment to the Canon, records token-level pathways, prevents dimensional and cross-scope duplication, protects uncertainty, and tests whether the measurement architecture genuinely improves prediction, protection, and flourishing. It strengthens RLS without creating an eighth dimension or a sixth gate.

21. Version history and change control

Version	Date	Status	Material changes
v1.0	July 2026	Superseded	Initial public research-source protocol.
v1.1	July 2026	Superseded by v1.2	Added Canon-bound scope redundancy treatment, PCC attachment, SGP measurement-standing separation, token-level pathways, machine-token namespaces, refined scalar-validity rule, defined incremental-value metrics, research-ethics routing, expanded conformance vectors, and verified package artifacts.

Version	Date	Status	Material changes
v1.2	July 2026	Superseded	Added dimension-set versioning and migration, provisional boundary-reliability triggers, allocate-versus-split and allocation-residual controls, tuple crosswalk, psychometric method options, configural-metric-scalar invariance ladder, time-window coherence, severe subgroup-harm visibility, weight sensitivity, named comparators, a runnable validation-study companion, seven new conformance vectors, and core-integration candidate materials.
v1.3	July 2026	Superseded	Registered WDBIP beneath RLS, synchronized run-record v2 and maturity interfaces, and removed candidate-only integration language.
v1.3.1	July 2026	Superseded	Repaired release identity, title structure, and package integration references.
v1.4	July 2026	Current	Synchronized the v1_4 schema, validator, example, vectors, references, version history, PCC terminology, and current MathGov component pins while preserving measurement and boundary rules.

Change control rule: future revisions must state whether they alter normative classification rules, machine-schema semantics, validation hypotheses, examples, or only editorial presentation. A release must not claim WDBIP conformance until the exact DOCX, PDF, Markdown, schema, example, validator, test vectors, validation-study companion, dimension-migration companion, manifest, and checksum ledger are synchronized. Reciprocal MathGov core integration additionally requires matching Canon, Component Map, Source Hierarchy, and package-manifest registration.

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Appendix A. Machine-readable companion

The official v1.4 release bundle includes wdbip_record_v1_4.schema.json, validate_wdbip_v1_4.py, WDBIP_VALIDATION_STUDY_PROTOCOL_v1.0.md, WDBIP_DIMENSION_SET_MIGRATION_STANDARD_v1.0.md, and the positive and negative conformance vectors. The schema encodes protocol identity, Canon and PCC linkage, the canonical seven dimensions, Union Scope treatment, record statuses, field-namespaced pathway and evidence tokens, and the following invariant rules:

Invariant	Value
Protocol role	Welfare measurement standard beneath RLS
Adds a gate	False
Adds a dimension	False
Unknown equals zero	False
Metaphysical truth is a welfare measure	False
Canonical dimensions	D1 Material through D7 Environment
Governing conflict rule	RippleLogic Canon controls
Scope redundancy methods	EMERGENT_SCALE_JUSTIFICATION, DEDUPLICATION, ALLOCATION

Invariant	Value
PCC attachment	Required for claim-bearing Tier 2 and Tier 3 use
Full-scope record	Declares all seven canonical Union Scopes
Reduced-scope record	Requires a PCC Scope Coverage Declaration reference
NOT_MATERIAL use	Requires rationale and reopen trigger
Measurement establishes SGP standing	False
Welfare dimension set	WELFARE_DIMENSION_SET_7D_V1
Silent dimension migration	False
Unexplained allocation residual	False
Unmodeled time-window mismatch	False
Severe subgroup harm may be averaged away	False
Weight-sensitive result may be presented as uniquely determined	False

The schema and validator are record-validity surfaces, not empirical validation of welfare constructs, causal pathways, moral standing, or decision correctness.

Appendix B. Framework registration entry

Recommended Core Component Map entry:

> **Welfare Dimension Boundary and Interaction Protocol v1.4 (WDBIP) - Canon-subordinate companion welfare-measurement and record-conformance protocol beneath RLS. Governs primary dimension assignment, token-level interaction records, dependence and cross-scope duplication disclosure, uncertainty, PCC attachment, and validation posture. Does not add a gate, dimension, authority, or moral-status determination.**

Recommended source-hierarchy position: beneath the RippleLogic Canon and alongside RLS measurement and validation companions; linked to SGP where bearer protection or welfare-inclusion hypotheses are decision-material.

Appendix C. Compact reader glossary and ownership pointers

This table improves standalone readability without redefining objects owned elsewhere.

Term	Compact reading	Controlling owner
RG	Reality Grounding: claim, evidence, category, uncertainty, and physical/causal admissibility discipline	RippleLogic Canon
RF/NCRC	Rights Floor / Non-Compensatory Rights Constraint	RippleLogic Canon
TRC	Tail-Risk Constraint for catastrophic and irreversible loss	RippleLogic Canon
CSV	Containment and Structural Viability gate	RippleLogic Canon

Term	Compact reading	Controlling owner
RLS	Residual welfare ranking among selectable options	RippleLogic Canon
PCC	Provenance and Compliance Certificate for the run	RippleLogic Canon
SGP	Sentience Gradient Protocol	SGP
FPP	Full Protection Plateau	SGP
AIU	All-Encompassing Infinite Union doctrine; not established by a welfare score	AIU-UBE foundational treatise / RG claim boundary
Effect token	Bounded claim about one state, bearer, baseline, option, and time window	Canon definition consumed by WDBIP
WDBIP profile	Structured welfare evidence record W(u,d,t)	WDBIP
WDBIP pathway	Token-level interaction record I(s -> t)	WDBIP
Weight sensitivity	A material change in ranking, decisiveness, or subgroup conclusion across plausible governed weights	Canon RLS/PLSS plus WDBIP disclosure

v1.3 integration delta (Lineage)

v1.3 synchronized WDBIP to the then-current Canon v12.3 line and contemporary SGP v8.x interface, registered it as a normative implementation companion beneath RLS, bound machine references to run-record v2 and the typed maturity register, and removed candidate-only integration language. It does not claim empirical validation of the seven dimensions or any pathway.